DEPARTMENT OF CHEMICAL & BIOMOLECULAR ENGINEERING
NORTH CAROLINA STATE UNIVERSITY

CHE 450 **Homework Set 5**

For ASPEN problems, submit the following to Moodle:

- *The saved ASPEN Plus files (file name: ASPEN_HW5_Name_Problem#)*
- *Include your answers as part of a PDF file submission to Gradescope containing for each problem:*
    1) *Screen capture of any 'Input' screens*
    2) *Screen capture of 'Stream Summary' section output showing data for all streams*
    3) *Final answer(s) to the problem*
    4) *Any additional screen captures requested in the problem statement*

Problem 1 (14%) 10,000 kg/hr water at 2 bar and 25°C enters a centrifugal pump, which discharges the water at a pressure of 10 bar. Use the STEAM-TA method and assume a pump efficiency of 70% along with a driver efficiency of 90%. How much electricity (in kW) is required to operate the pump?

Problem 2 (30%) 10,000 kg/hr water at 2 bar and 25°C enters a centrifugal pump. Use the STEAM-TA method and assume a pump efficiency of 70% along with a driver efficiency of 90%. You know that the pump curve describing operation of the pump follows this trend:

Head (m water)	Capacity (m^3/hr)
40	20
250	10
300	5
400	3

(a, 15%) How much electricity (in kW) is required to operate the pump? **(b, 15%)** What is the pressure (in bar) at the outlet of the pump?

Problem 3 (28%) 200 kmol/hr air at 1 atm and 25°C is isentropically compressed to a pressure of 3 bar. Use the IDEAL method and assume isentropic efficiency of 85%. **(a, 14%)** What is the required compressor horsepower (AKA brake horsepower) in kW? **(b, 14%)** What is the outlet temperature (not isentropic outlet temperature) of the air?

Problem 4 (14%) 10,000 kg/hr water at 25°C and 10 bar flows through a 1000 ft section of 2" SCH 40 carbon steel pipe. Use the STEAM-TA method. What is the pressure drop across the pipe (in bar)?

Problem 5 (14%) 10,000 kg/hr water at 25°C and 10 bar flows through a 1" metal-seated full-port ball valve intended to throttle the water pressure to 6 bar. Use the STEAM-TA method. What is the required valve % opening for this operation?